Utah State University

ECE 6010

Stochastic Processes, Fall 2025

Programming Exercise # 2

Due December 5, 2025 (last day of class)

The Kalman Filter

Introduction

The Kalman filter is a powerful tool used in a variety of applications in signal processing, controls, communications, etc. It follows in a fairly straightforward way from the basic principles we have been talking about in class.

In this programming exercise, you will be presented with the context in which the Kalman filter applies, and a derivation of it based on Bayesian probabilistic concepts. You will then program it in Matlab and test it on a particular system.

As you read through the derivation, perform those steps indicated by ☞, and ending with the symbol □. Turn in this work as part of the programming exercise.

Background

Let $\mathbf{X}$ and $\mathbf{Y}$ be Gaussian random vectors, with

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_x, \Sigma_x) \qquad \mathbf{Y} \sim \mathcal{N}(\boldsymbol{\mu}_y, \Sigma_y)$$

and, jointly,

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{bmatrix}$$

$$\operatorname{cov}(\mathbf{X}, \mathbf{Y}) = \begin{bmatrix} \Sigma_x & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_y \end{bmatrix} = \Sigma.$$

That is, $(\mathbf{X}, \mathbf{Y}) \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$.

Recall that conditioned upon a measurement of $\mathbf{Y} = \mathbf{y}$, $\mathbf{X}|(\mathbf{Y} = \mathbf{y})$ is still Gaussian:

$$\mathbf{X}|(\mathbf{Y} = \mathbf{y}) \sim \mathcal{N}(\boldsymbol{\mu}_{x|y}, \Sigma_{x|y}),$$

where

$$\boldsymbol{\mu}_{x|y} = \boldsymbol{\mu}_x + \Sigma_{xy}\Sigma_y^{-1}(\mathbf{y} - \boldsymbol{\mu}_y) \quad \text{and} \quad \Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}. \tag{1}$$

We learn this by simply writing out the definitions for the conditional densities:

$$\begin{aligned} f_{X|Y}(\mathbf{x}|\mathbf{Y} = \mathbf{y}) &= \frac{f_{XY}(\mathbf{x}, \mathbf{y})}{f_Y(\mathbf{y})} \\ &= f_X(\mathbf{x}) \frac{f_{Y|X}(\mathbf{y}|\mathbf{x})}{f_Y(\mathbf{y})} \end{aligned} \tag{2}$$

Here $f_X(\mathbf{x})$ is the prior probability, and $f_{Y|X}(\mathbf{y}|\mathbf{x})$ is the likelihood function. By writing out explicitly the densities and completing the square, the form of the solution can be obtained.

☞ (5 pts) **Bayesian updating in the scalar case:** Let X and Y be scalar random variables and let (X, Y) be jointly Gaussian distributed, where X and Y are scalar random variables, with

$$E\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$$

$$\operatorname{cov}(X, Y) = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{yx} & \sigma_y^2 \end{bmatrix}.$$

Show by direction substitution in (2) that

$$X|(Y=y) \sim \mathcal{N}(\mu_{x|y}, \sigma^2_{x|y})$$

with

$$\mu_{x|y} = \mu_x + \frac{\sigma_{xy}}{\sigma_y^2}(y - \mu_y).$$

and

$$\sigma^2_{x|y} = \sigma^2_x - \sigma^2_{xy}/\sigma^2_y.$$

Hint: Complete the square.

□

The State Variable System

We are given a linear (possibly time-varying) system description in discrete time state variable form. The state update equation is

$$\boxed{\mathbf{x}_{t+1} = A_t\mathbf{x}_t + \mathbf{w}_t.} \tag{3}$$

Here, A_t is the state update matrix, and $\mathbf{w}_t$ is the input signal. For our purposes, we assume each vector $\mathbf{w}_t$ to be Gaussian, with mean $\mathbf{0}$ and covariance Q_t:

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t).$$

Furthermore, we assume that the sequence $\mathbf{w}_t$ is *white*:

$$E[\mathbf{w}_t\mathbf{w}_\tau^T] = 0 \text{ if } t \neq \tau.$$

Think of (3) as describing the way a system moves that has noise as its input. That is, the system is moving around under the buffetings of noise. We refer to $\mathbf{w}_t$ as the input noise.

There is also an observation equation,

$$\boxed{\mathbf{y}_t = C_t\mathbf{x}_t + \boldsymbol{\nu}_t.} \tag{4}$$

Here C_t is the observation matrix, and $\boldsymbol{\nu}_t$ is observation noise. We make observations of (part of) $\mathbf{x}_t$, but obtain only a noisy view of the true values. We refer to $\boldsymbol{\nu}_t$ as the observation noise. We assume that $\boldsymbol{\nu}_t$ is a Gaussian random vector with zero mean and covariance R_t:

$$\boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, R_t).$$

We furthermore assume that the sequence $\boldsymbol{\nu}_t$ is white:

$$E[\boldsymbol{\nu}_t\boldsymbol{\nu}_\tau^T] = 0 \text{ if } t \neq \tau.$$

We furthermore assume (merely for simplicity) that the observation noise and the system noise are uncorrelated:

$$E[\mathbf{w}_t\boldsymbol{\nu}_\tau^T] = 0 \text{ for all } t, \tau.$$

The Kalman filter problem is this: Given a sequence of observations, $\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_t$, determine an estimate for the state $\mathbf{x}_t$ at time t, using all the observations up to time t. What complicates this, of course, is that the system is moving around at random, and we only have noisy observations. On the other hand, the problem is simplified by two key facts:

1. The noise is Gaussian, hence the state variable $\mathbf{x}_t$ is Gaussian.
2. The system is Markov. That is, if I tell you what the state is at time t, $\mathbf{x}_t$, then it doesn't matter how you arrived at that state, or even what you observe in that state. Everything that follows after this depends on knowing $\mathbf{x}_t$.

We use the symbol $\mathcal{Y}_t$ to represent the sequence of observations up to time t:

$$\mathcal{Y}_t = \{\mathbf{y}_0, \mathbf{y}_1, \ldots, \mathbf{y}_t\}.$$

Given these assumptions and notation, we observe that the state $\mathbf{x}_t$, conditioned on the sequence of observations $\mathbf{y}_t$, must also be Gaussian. We denote the mean of this conditional Gaussian as $\hat{\mathbf{x}}_{t|t}$. It is an estimate (hence the hat) of $\mathbf{x}_t$, given observations up to time t, hence the $|t$. We denote the covariance of this conditional Gaussian as $P_{t|t}$. We can thus write

$$\mathbf{x}_t|\mathcal{Y}_t \sim \mathcal{N}(\hat{\mathbf{x}}_{t|t}, P_{t|t}).$$

We proceed on the estimation problem as follows. We assume that we have a distribution for $\mathbf{x}_t|\mathcal{Y}_{t-1}$. That is, we know $\hat{\mathbf{x}}_{t|t-1}$ and $P_{t|t-1}$. Now an observation $\mathbf{y}_t$ is measured, and we need to incorporate this new observation in. We have

$$\mathcal{Y}_t = \{\mathbf{y}_0, \ldots, \mathbf{y}_{t-1}, \mathbf{y}_t\} = \{\mathcal{Y}_{t-1}, \mathbf{y}_t\}.$$

The new state, now conditioned on the observations $\mathcal{Y}_t$, is Gaussian distributed, with mean and covariance we denote as $\hat{\mathbf{x}}_{t|t}$ and $P_{t|t}$. That is,

$$\mathbf{x}_t|\mathcal{Y}_t \sim \mathcal{N}(\hat{\mathbf{x}}_{t|t}, P_{t|t}).$$

Now the system advances in time proceeding to a new state $\mathbf{x}_{t+1}$, but we haven't made a new observation yet. The estimate of the new state, $\mathbf{x}_{t+1}$, conditioned on the observations $\mathcal{Y}_t$, is also Gaussian distributed, with mean and covariance we denote as $\hat{\mathbf{x}}_{t+1|t}$ and $P_{t+1|t}$. That is,

$$\mathbf{x}_{t+1}|\mathcal{Y}_t \sim \mathcal{N}(\hat{\mathbf{x}}_{t+1|t}, P_{t+1|t})$$

To summarize: Starting from the density $\mathcal{N}(\hat{\mathbf{x}}_{t|t-1}, P_{t|t-1})$:

1. "Update" this density by incorporating a new observation at time t, by computing the mean and the covariance $\hat{\mathbf{x}}_{t|t}$ and $P_{t|t}$.
2. "Propagate" the density desribed by $(\hat{x}_{t|t}, P_{t|t})$ to the new state $\mathbf{x}_{t+1}$ at time $t+1$ (but do not incorporate new observations yet) by computing the mean and covariance $\hat{\mathbf{x}}_{t+1|t}$ and $P_{t+1|t}$.

By these two steps, the parameters $(\hat{x}_{t|t}, P_{t|t})$ are transformed into $(x_{t+1|t+1}, P_{t+1|t+1})$, following which the whole process can be repeated with $t \leftarrow t+1$.

We now consider how to perform these propagation and update steps, starting first with the update step.

The Update Step

For the update step, we desire to compute $f(\mathbf{x}_t|\mathcal{Y}_t)$, based on knowing $f(\mathbf{x}_t|\mathcal{Y}_{t-1})$. This can be obtained as follows:

$$\boxed{f(\mathbf{x}_t|\mathcal{Y}_t) = f(\mathbf{x}_t|\mathcal{Y}_{t-1})\frac{f(\mathbf{y}_t|\mathbf{x}_t)}{f(\mathbf{y}_t|\mathcal{Y}_{t-1})}.} \tag{5}$$

☞ (5 pts) Justify (5). Hint: use conditional independence.

□

The fact that everything in sight in (5) is Gaussian suggests that the new density $f(\mathbf{x}_t|\mathcal{Y}_t)$ is also going to be Gaussian. This could be verified by writing down the densities and completing the square. But compare (5) with (2). Except for the fact that in (5) everything in sight is conditioned upon $\mathcal{Y}_{t-1}$, the form of the equations is exactly the same, with $f(\mathbf{x}_t|\mathcal{Y}_{t-1})$ acting as the prior probability, and $f(\mathbf{y}_t|\mathbf{x}_t) = f(\mathbf{y}_t|\mathbf{x}_t, \mathcal{Y}_{t-1})$ acting as the likelihood. We can therefore write down the parameters of the distribution $f(\mathbf{x}_t|\mathcal{Y}_t)$ by analogy with (1). We thus have

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + R_{xy|\mathcal{Y}_{t-1}} R^{-1}_{y|\mathcal{Y}_{t-1}} (\mathbf{y}_t - E[\mathbf{y}_t|\mathcal{Y}_{t-1}]) \tag{6}$$

and

$$P_{t|t} = P_{t|t-1} - R_{xy|\mathcal{Y}_{t-1}} R_{y|\mathcal{Y}_{t-1}} R^T_{xy|\mathcal{Y}_{t-1}}, \tag{7}$$

where:

1. $R_{xy|\mathcal{Y}_{t-1}}$ denotes the covariance of $\mathbf{x}_t$ and $\mathbf{y}_t$, conditioned upon $\mathcal{Y}_{t-1}$,

$$R_{xy|\mathcal{Y}_{t-1}} = E[(\mathbf{x}_t - E[\mathbf{x}_t])(\mathbf{y}_t - E[\mathbf{y}_t])^T|\mathcal{Y}_{t-1}]$$

2. $R_{y|\mathcal{Y}_{t-1}}$ denotes the covariance of $\mathbf{y}_t$, conditioned upon $\mathcal{Y}_{t-1}$,

$$R_{y|\mathcal{Y}_{t-1}} = E[(\mathbf{y}_t - E[\mathbf{y}_t])(\mathbf{y}_t - E[\mathbf{y}_t])^T|\mathcal{Y}_{t-1}].$$

(Stop here and think about this until you understand it. This is important!)

It remains now to find the quantities $E[\mathbf{y}_t|\mathcal{Y}_{t-1}]$, $R_{xy|\mathcal{Y}_{t-1}}$, and $R_{y|\mathcal{Y}_{t-1}}$.

- $E[\mathbf{y}_t|\mathcal{Y}_{t-1}]$.

$$E[\mathbf{y}_t|\mathcal{Y}_{t-1}] = E[C_t\mathbf{x}_t + \boldsymbol{\nu}_t|\mathcal{Y}_{t-1}] = C_tE[\mathbf{x}_t|\mathcal{Y}_{t-1}] + E[\boldsymbol{\nu}_t|\mathcal{Y}_{t-1}]$$

  The conditional expectation $E[\mathbf{x}_t|\mathcal{Y}_{t-1}]$ is what we have denoted by $\hat{\mathbf{x}}_{t|t-1}$ (think about it!). The noise $\boldsymbol{\nu}_t$ at time t is uncorrelated with the noise in the observations at any previous time, and the noise is zero mean, so $E[\boldsymbol{\nu}_t|\mathcal{Y}_{t-1}] = 0$. Hence

$$E[\mathbf{y}_t|\mathcal{Y}_{t-1}] = C_t\hat{\mathbf{x}}_{t|t-1}.$$

- $R_{xy|\mathcal{Y}_{t-1}}$.

$$\begin{aligned}
R_{xy|\mathcal{Y}_{t-1}} &= E[(\mathbf{x}_t - E[\mathbf{x}_t])(\mathbf{y}_t - E[\mathbf{y}_t])^T|\mathcal{Y}_{t-1}] \\
&= E[(\mathbf{x}_t - E[\mathbf{x}_t])(C_t\mathbf{x}_t + \boldsymbol{\nu}_t - C_t\hat{\mathbf{x}}_{t|t-1})^T|\mathcal{Y}_{t-1}] \qquad \text{(using definition and previous result)} \\
&= E[(\mathbf{x}_t - E[\mathbf{x}_t])(C_t(\mathbf{x}_t - \hat{\mathbf{x}}_{t|t-1}) + \boldsymbol{\nu}_t)^T|\mathcal{Y}_{t-1}] \\
&= E[(\mathbf{x}_t - E[\mathbf{x}_t])(\mathbf{x}_t - \hat{\mathbf{x}}_{t|t})^T|\mathcal{Y}_{t-1}]C_t^T + E[(\mathbf{x}_t - E[\mathbf{x}_t])\boldsymbol{\nu}_t^T|\mathcal{Y}_{t-1}] \\
&= P_{t|t-1}C_t^T + 0 \qquad \text{(using (11))}
\end{aligned}$$

- $R_{y|\mathcal{Y}_{t-1}}$

$$\begin{aligned}
R_{y|\mathcal{Y}_{t-1}} &= E[(\mathbf{y}_t - E[\mathbf{y}_t])(\mathbf{y}_t - E[\mathbf{y}_t])^T|\mathcal{Y}_{t-1}] \\
&= E[(C_t(\mathbf{x}_t - \hat{\mathbf{x}}_{t|t-1}) + \boldsymbol{\nu}_t)((\mathbf{x}_t - \hat{\mathbf{x}}_{t|t-1})^TC_t^T + \boldsymbol{\nu}_t^T)|\mathcal{Y}_{t-1}] \\
&= C_tP_{t|t-1}C_t^T + R_t \qquad \text{(using (11))}
\end{aligned}$$

Putting these pieces together into (6) we have the following mean update step:

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + P_{t|t-1}C_t^T(C_tP_{t|t-1}C_t^T + R_t)^{-1}(\mathbf{y}_t - C_t\hat{\mathbf{x}}_{t|t-1})$$

At this point it is convenient to introduce the *Kalman gain*:

$$K_t = P_{t|t-1}C_t^T(C_tP_{t|t-1}C_t^T + R_t)^{-1}.$$

Then we can write the mean update as

$$\boxed{\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t(\mathbf{y}_t - C_t\hat{\mathbf{x}}_{t|t-1}).}$$

From (7), the update step for the covariance is

$$P_{t|t} = P_{t|t-1} - P_{t|t-1}C_t^T(C_tP_{t|t-1}C_t^T)^{-1}C_tP_{t|t-1},$$

or

$$\boxed{P_{t|t} = (I - K_tC_t)P_{t|t-1}.}$$

The Kalman filter is initialized starting from some initial Gaussian distribution with mean $\mathbf{x}_{0|-1}$ and covariance $P_{0|-1}$.

The Propagation Step

Let $f(\mathbf{x}_t|\mathcal{Y}_t)$ denote the conditional density of $\mathbf{x}_t$ at time t, given the observations up to time t. In the propagation step, we desire to obtain an expression for $f(\mathbf{x}_{t+1}|\mathcal{Y}_t)$ — a description of the state at the next time, using observations up to time t.

The propagate step can be expressed using the law of total probability and the definition of conditional probabilities as follows. Starting from $f(\mathbf{x}_{t+1}|\mathbf{x}_t, \mathcal{Y}_t)$, then

$$f(\mathbf{x}_{t+1}|\mathbf{x}_t, \mathcal{Y}_t)f(\mathbf{x}_t|\mathcal{Y}_t) = f(\mathbf{x}_{t+1}, \mathbf{x}_t|\mathcal{Y}_t).$$

Then the desired density can be obtained by integrating out (marginalizing) the variable $\mathbf{x}_t$:

$$f(\mathbf{x}_{t+1}|\mathcal{Y}_t) = \int f(\mathbf{x}_{t+1}, \mathbf{x}_t|\mathcal{Y}_t)\, d\mathbf{x}_t = \int f(\mathbf{x}_{t+1}|\mathbf{x}_t, \mathcal{Y}_t)f(\mathbf{x}_t|\mathcal{Y}_t)\, d\mathbf{x}_t.$$

To compute this expression, we need to know

$$f(\mathbf{x}_{t+1}|\mathbf{x}_t, \mathcal{Y}_t).$$

Here, however, is where the Markovity, or the state-space structure, of the problem enters in. If I tell you $\mathbf{x}_t$, and you want to determine $\mathbf{x}_{t+1}$, then the information in $\mathcal{Y}_t$ has no bearing on $\mathbf{x}_{t+1}$. That is, $\mathbf{x}_{t+1}$ is **conditionally independent** of $\mathcal{Y}_t$, given that $\mathbf{x}_t$ is also known. Hence we can write

$$f(\mathbf{x}_{t+1}|\mathbf{x}_t, \mathcal{Y}_t) = f(\mathbf{x}_{t+1}|\mathbf{x}_t).$$

The propagation equation thus becomes

$$\boxed{f(\mathbf{x}_{t+1}|\mathcal{Y}_t) = \int f(\mathbf{x}_{t+1}|\mathbf{x}_t)f(\mathbf{x}_t|\mathcal{Y}_t)\, d\mathbf{x}_t.} \tag{8}$$

We now must come up with an explicit expression for $f(\mathbf{x}_{t+1}|\mathbf{x}_t)$.

Consider the state equation

$$\mathbf{x}_{t+1} = A_t\mathbf{x}_t + \mathbf{w}_t.$$

Since $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t)$, if I tell you $\mathbf{x}_t$, then $\mathbf{x}_{t+1}$ simply represents a shift of $\mathbf{w}_t$, which changes its mean but not its covariance. Thus

$$\mathbf{x}_{t+1}|\mathbf{x}_t \sim \mathcal{N}(A_t\mathbf{x}_t, Q_t).$$

This is important: think about this until you understand it! Another way of thinking about this is that the behavior of $\mathbf{x}_{t+1}$, given $\mathbf{x}_t$, in the mean simply acts like the system: we propagate $\mathbf{x}_t$ through the system to obtain the new state $A_t\mathbf{x}_t$.

Given our inductive statement, we have an expression for $f(\mathbf{x}_t|\mathcal{Y}_t)$. Now we also have an expression for $f(\mathbf{x}_{t+1}|\mathbf{x}_t)$. Let us use the notation $\mathcal{N}(\mathbf{x}_{t+1}; A_t\mathbf{x}_t, Q_t)$ as a shorthand for the density $f(\mathbf{x}_{t+1}|\mathbf{x}_t)$, and $\mathcal{N}(\mathbf{x}_t, \hat{\mathbf{x}}_{t|t}, P_{t|t})$ as a shorthand for the density of $\mathbf{x}|\mathcal{Y}_t$.

We can now write the propagation equation as

$$\begin{aligned} f(\mathbf{x}_{t+1}|\mathcal{Y}_t) &= \int f(\mathbf{x}_{t+1}|\mathbf{x}_t)f(\mathbf{x}_t|\mathcal{Y}_t)\, d\mathbf{x}_t \qquad (9)\\ &= \int \mathcal{N}(\mathbf{x}_{t+1}; A_t\mathbf{x}_t, Q_t)\mathcal{N}(\mathbf{x}_t; \hat{\mathbf{x}}_{t|t}, P_{t|t})\, d\mathbf{x}_t \\ &= C\int \exp\left[-\frac{1}{2}(\mathbf{x}_{t+1} - A_t\mathbf{x}_t)^T Q_t^{-1}(\mathbf{x}_{t+1} - A_t\mathbf{x}_t)\right] \exp\left[-\frac{1}{2}(\mathbf{x}_t - \hat{\mathbf{x}}_{t|t})^T P_{t|t}^{-1}(\mathbf{x}_t - \hat{\mathbf{x}}_{t|t})\right] d\mathbf{x}_t, \qquad (10)\end{aligned}$$

where C is the constant resulting from the two pdfs. Performing the integration (which involves completing the square), we find that $\mathbf{x}_{t+1}|\mathcal{Y}_t$ is Gaussian with mean

$$\hat{\mathbf{x}}_{t+1|t} = A_t\hat{\mathbf{x}}_{t|t}$$

and covariance

$$P_{t+1|t} = A_tP_{t|t}A_t^T + Q_t.$$

The mean behaves just as if we ran the state $\hat{\mathbf{x}}_{t|t}$ through the system, without the noise. The covariance gets "bigger", adding on an extra term of variance with Q_t.

☞ (5 pts) **Propagation in the scalar case**. Let x_t be a scalar variable, with system

$$x_{t+1} = a_t x_t + w_t.$$

Let $x_{t+1}|x_t \sim \mathcal{N}(x_{t+1}; a_t x_t, Q_t)$ and $x_t|\mathcal{Y}_t \sim \mathcal{N}(x_t; \hat{x}_{t|t}, P_t)$. Perform the integration in (10) to show that $x_{t+1}|\mathcal{Y}_t \sim \mathcal{N}(a_t\hat{x}_{t|t}, a_t^2 P_{t|t} + Q_t)$.

□

We make an observation about the notation. Since $P_{t+1|t}$ is a covariance of $\mathbf{x}_{t+1}|\mathcal{Y}_t$, we have, by definition,

$$P_{t+1|t} = E[(\mathbf{x}_{t+1} - E[\mathbf{x}_{t+1}])(\mathbf{x}_{t+1} - E[\mathbf{x}_{t+1}])^T|\mathcal{Y}_t] = E[(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})^T|\mathcal{Y}_t]. \quad (11)$$

Summary of the Kalman filter

We summarize as follows. For the state update equation

$$\mathbf{x}_{t+1} = A_t\mathbf{x}_t + \mathbf{w}_t$$

and observation equation

$$\mathbf{y}_t = C_t\mathbf{x}_t + \boldsymbol{\nu}_t$$

where

$$\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, Q_t) \qquad \boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, R_t) \qquad E[\mathbf{w}_t\mathbf{w}_\tau^T] = 0 \text{ if } t \neq \tau :$$

Start with the initial condition is $\hat{\mathbf{x}}_{0|-1}$ (the state at time 0, with no previous observations) and $P_{0|-1}$ (the initial covariance). For each observation $\mathbf{y}_t, t = 0, 1, \ldots$, the Kalman filter follows the following steps, starting at $t = 0$:

1. **Compute the Kalman gain**: $K_t = P_{t|t-1}C_t^T(C_tP_{t|t-1}C_t^T + R_t)^{-1}$
2. **State estimate update**: $\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t(\mathbf{y}_t - C_t\hat{\mathbf{x}}_{t|t-1})$.
3. **Error covariance update**: $P_{t|t} = (I - K_tC_t)P_{t|t-1}$.
4. **State estimation propagation**: $\hat{\mathbf{x}}_{t+1|t} = A_t\hat{\mathbf{x}}_{t|t}$
5. **Error covariance propagation**: $P_{t+1|t} = A_tP_{t|t}A_t^T + Q_t$.

There are many other variations and considerations, such as the following:

1. What if the system is nonlinear? What if the noise is nongaussian?
2. How to minimize the computations? Is there a way of computing in a numerically stable way?
3. What about continuous time systems?

There are answers to all of these questions in the literature. However, our derivation suffices for our introductory purposes.

A Sample System

The motion of a particle of mass m acted upon by a net force $f(t)$ can be described by the differential equation

$$f(t) = m\ddot{z}(t),$$

where $z(t)$ is the position of the particle at time t. We express this second-order differential equation in state variable form as follows. Let $x_1(t) = z(t)$ be the position variable, and $x_2(t) = \dot{z}(t) = \dot{x}_1(t)$ be the velocity variable. Then we can write this second order system as

$$\begin{bmatrix}\dot{x}_1(t)\\ \dot{x}_2(t)\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}x_1(t)\\ x_2(t)\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix} f.$$

Let

$$\tilde{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

denote the system matrix in continuous time and let

$$\tilde{\mathbf{b}} = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$$

denote the input matrix. This differential equation can be approximated as a discrete-time system as follows. Write the differential equation as a difference equation:

$$\frac{\begin{bmatrix} x_1(t+h) \\ x_2(t+h) \end{bmatrix} - \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}}{h} = \tilde{A} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \tilde{\mathbf{b}} f,$$

where h is small, and solve for the most recent update step:

$$\begin{bmatrix} x_1(t+h) \\ x_2(t+h) \end{bmatrix} = (I + h\tilde{A}) \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + h\tilde{\mathbf{b}} f$$

This we write in a discrete-time notation as

$$\mathbf{x}_{t+1} = (I + h\tilde{A})\mathbf{x}_t + h\tilde{\mathbf{b}} f$$

where

$$\mathbf{x}_t = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \qquad \mathbf{x}_{t+1} = \begin{bmatrix} x_1(t+h) \\ x_2(t+h) \end{bmatrix}.$$

Let

$$A = I + h\tilde{A} = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}$$

and $\mathbf{b} = h\tilde{\mathbf{b}}$. We obtain

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + \mathbf{b} f$$

Thus, a single dimensional mechanical system gives rise to a two-dimensional discrete-time state space system.

Let us suppose now that there are two position coordinates, $z_1(t)$ and $z_2(t)$, described by the equations

$$f_1 = m\ddot{z}_1 \qquad f_2 = m\ddot{z}_2$$

Now introduce state variables

$$x_1 = z_1 \quad x_2 = \dot{z}_1 \quad x_3 = z_2 \quad x_4 = \dot{z}_2$$

Then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$$

Discretizing as before, we can write

$$\mathbf{x}_{t+1} = A\mathbf{x}_t + B\mathbf{f} \tag{12}$$

where

$$A = \begin{bmatrix} 1 & h & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & h \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 0 \\ \frac{h}{m} & 0 \\ 0 & 0 \\ 0 & \frac{h}{m} \end{bmatrix}$$

Now let $\mathbf{w}_t = B\mathbf{f}$ and suppose that $\mathbf{w}_t$ is a Gaussian random vector. That is, the particle is driven about by random forces.

Suppose that in this system we are only able to measure the position variables, in the presence of noise. Let

$$\mathbf{y}_t = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}$$

denote the measurement vector. Then

$$\mathbf{y}_t = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}_t + \boldsymbol{\nu}_t. \tag{13}$$

Suppose that $\boldsymbol{\nu}_t \sim \mathcal{N}(\mathbf{0}, R_t)$.

With these equations we now have a complete state space description of the system.

Programming Assignment

☞ (5 pts) In Matlab or Python or Julia, program a function `kalman` which performs Kalman filtering. The function declaration should be something like the following.

```
function [xhat,P] = kalman(y,xtt,Ptt,A,C,Q,R)
%
% Computes the Kalman filter esimate xhat(t+1|t+1)
% for the system x(t+1) = Ax(t) + w
%                y(t) = Cx(t) + v
% where cov(w) = Q  and cov(v) = R,
% The prior estimate is x(t|t) = xtt, and the prior covariance is Ptt.
%
% (By passing in (A,B,C,Q,R) each time, the possibility of a time-varying
% system is accomodated, at the expense of some computational efficiency.)
```

Use $h = 0.01$.

☞ (5 pts) Test your function on the four-dimensional dynamical system described in (12) and (13). Let

$$\mathrm{cov}(\mathbf{w}_t) = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 2 & 3 \end{bmatrix}$$

and

$$\mathrm{cov}(\boldsymbol{\nu}_t) = 3I.$$

Make a plot that compares the true state (as computed by (12) with the estimated state, as computed by your Kalman filter. You may want to plot (x_1, x_3) (the position) on one set of axes and (x_2, x_4) on another set of axes. (To simulate the system, you will need to generate the Gaussian noise vectors $\mathbf{w}_t$ and $\boldsymbol{\nu}_t$, with the stated covariance matrices.)

Assume that $\mathbf{x}_{-1|0}$ is equal to $(0, 0, 0, 0)$, and $P_{-1|0} = \mathrm{cov}(\mathbf{w}_t)$.

☞ (5 pts) Try different values of system and observation covariance matrices, making plots of true *vs.* estimated states in each case. Comment on how well the Kalman filter is able to track the true state.

☞ (5 pts) Turn in the exercises, your code, your plots, the results of the Kalman filter simulations, and observations about what you have learned.

□